

Free License Give-Away — Interview Task Brief

Format: System Design + Coding · **Total time:** 120 minutes · **Coding language:** C#

Scenario

A company is running a free software give-away campaign. The software is in high demand, and only a limited number of licenses are available.

- There are exactly **1,000 licenses** available in total.
 - Thousands of people are expected to apply at the same time, as soon as applications open.
 - Assume a large burst of traffic around the opening moment (tens of thousands of concurrent requests), tapering off quickly afterward. You don't need to design for sustained high traffic beyond that initial spike.
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Part A — System Design

Design a system (architecture diagram plus explanation) that satisfies **all** of the following:

1. A frontend accepts an email address and a phone number as a license application.
2. Exactly 1,000 licenses must be issued — not one fewer, not one more.
3. Each person may receive only one license. The same email address or the same phone number must not be able to receive multiple licenses.
4. Each license is delivered as a PDF containing a license code and a certificate. This PDF must remain secret, accessible only to the license owner and the company.
5. The process must be fair: wherever possible, the first person to apply (“click”) should be the first person served.
6. The applicant must receive feedback in the UI once their PDF is ready, and be able to retrieve it.
7. The website must remain highly available and low-latency at all times, including during the traffic spike.

This part is a diagram plus a verbal/written explanation — you do not need to write code, build a real frontend, or provision any actual infrastructure. You're welcome to use any drawing or diagramming tool you're comfortable with; a quick, roughly-drafted sketch is perfectly fine, as long as it communicates your design clearly.

Part B — Coding Exercise

Implement the API endpoint(s) that accept a license application, as a rough, representative implementation — not a production-grade one.

- **Language: C#** (ASP.NET Core minimal API or controller-based — either is fine).
 - AI coding tools (Copilot, ChatGPT, Claude, etc.) are allowed. Be ready to explain and justify your code choices afterward.
 - The database can be ignored entirely — use an in-memory variable (e.g., a `List`, `Dictionary`, or `HashSet`) to represent stored applications/licenses.
 - Scope is just the request-handling logic for this one endpoint — you don't need to build or wire up any real queue, background worker, cloud service, or actual PDF generation. Stub those out (e.g., a function that logs or returns a placeholder).
 - Focus on getting the request/response shape and the core logic right. It doesn't need to be a runnable, polished project — a code file you can talk through on screen is enough.
 - It's fine to leave parts as comments, TODOs, or simplified stand-ins, as long as you can explain what a real version would do.
 - It's also fine not to finish. We're more interested in your approach and reasoning than in a complete, polished result — don't spend your remaining time gold-plating one part at the expense of the other.
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How the Session Will Run

- **Introduction (10 min):** We'll walk through this brief together and answer any clarifying questions.
- **Solo work (60 min):** You'll work alone and unobserved on your own computer — we'll leave the call and rejoin at the end. You manage your own time split between Part A and Part B; there's no enforced sub-timer.
- **Review (50 min):** We'll rejoin, and you'll walk us through your design and your code. Expect follow-up questions on both.

A few logistics notes:

- You'll be sharing your screen during the review. We don't need your design or code submitted to us separately — this is a live walkthrough, not an async review.
 - The call will not be recorded. The facilitator may occasionally take a screenshot for internal notes.
 - If you have dyslexia, ADHD, or any other condition that means you'd benefit from more time, let us know when we're scheduling the session — we're happy to arrange a longer slot.
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What to Have Ready by the End

- A system design (diagram + explanation) covering Part A.

- A rough code sketch covering Part B — doesn't need to run, just needs to be something you can walk us through.